

Learning goals

- Compute Fourier coefficients of engine signals analytically and numerically; understand the line spectrum of the 60-2 crank signal and the effect of the missing teeth.
- Relate the decay time of a resonance to the width of its spectral peak (Lorentzian line shape, bandwidth, quality factor).
- Evaluate the DTFT by direct summation and use its properties (periodicity, symmetry).
- Use *engine orders* to distinguish speed-proportional components from fixed-frequency resonances; track orders during a speed change.

Background

A signal with period P has the Fourier series $x(\theta) = \sum_k c_k e^{j2\pi k\theta/P}$, $c_k = \frac{1}{P} \int_0^P x(\theta) e^{-j2\pi k\theta/P} d\theta$. For crank-angle signals we choose $P = 360^\circ$: then k counts oscillations per crankshaft revolution and is called the *engine order*. At speed n (rpm) order k has the frequency $f = k n/60$. Aperiodic signals have a Fourier transform $X(f) = \int x(t) e^{-j2\pi f t} dt$; a sampled signal $x[n] = x(nT_s)$ has the DTFT

$$X(e^{j\Omega}) = \sum_n x[n] e^{-j\Omega n}, \quad \Omega = 2\pi f/f_s,$$

which is 2π -periodic in Ω (f_s -periodic in f) and conjugate symmetric for real $x[n]$. For a finite record of N samples it can be evaluated at any frequency by direct summation; the FFT (Sheet 8) is only a fast way to do this on a regular grid. A sine of amplitude A whose frequency fits an integer number of periods into the record has $|X| = AN/2$, so $A = 2|X|/N$.

Exercise 6.1 Fourier series and transforms of engine signals**PAPER**

- a) A complete 60-tooth wheel gives a rectangular signal with period 6° and duty cycle 50%. Compute its Fourier coefficients with respect to $P = 360^\circ$. Which orders occur, what are c_0 , $|c_{60}|$, $|c_{120}|$, $|c_{180}|$? Which frequency does order 60 have at 3000 rpm?
- b) For the 60-2 wheel (tooth $m = 0, \dots, 57$ HIGH on $[6m, 6m + 3)^\circ$) show that

$$|c_k| = \frac{3}{360} \left| \operatorname{sinc} \frac{k}{120} \right| \left| \frac{\sin(29\pi k/30)}{\sin(\pi k/60)} \right| \quad (k \neq 60m), \quad |c_{60m}| = \frac{58}{120} \left| \operatorname{sinc} \frac{m}{2} \right|,$$

with $\operatorname{sinc} u = \sin(\pi u)/(\pi u)$. Evaluate c_0 , $|c_1|$, $|c_{59}|$, $|c_{60}|$, $|c_{61}|$ and explain the result as a “once-per-revolution modulation”. Hint: $x_{60-2} = x_{60} - x_{\text{gap}}$.

- c) Compute the Fourier transform of the knock model $x(t) = e^{-t/\tau} \sin(2\pi f_0 t)$, $t \geq 0$. Show that near f_0 the magnitude is a Lorentzian curve, and derive the -3 dB bandwidth Δf and the quality factor $Q = f_0/\Delta f$. Evaluate for the knock mode ($f_0 = 6.5$ kHz, $\tau = 0.8$ ms) and the valve impact (8 kHz, 0.4 ms). What follows for the bandwidth of a knock band-pass filter?
- d) Engine orders: list the frequencies of order 2 (combustions of the 4-cylinder engine), order 60 (crank teeth) and order 0.5 (camshaft) at 1500, 3000 and 6000 rpm, and the order at which the 6.5 kHz knock resonance appears at these speeds.

Solution

a) The signal is periodic with 6° , so only orders $k = 60m$ occur. For a rectangular pulse of width $w = 3^\circ$ in the period $P_0 = 6^\circ$: $c_{60m} = \frac{w}{P_0} \text{sinc}(mw/P_0) e^{-j\pi m w/P_0} = \frac{1}{2} \text{sinc} \frac{m}{2} e^{-j\pi m/2}$. $c_0 = 0.5$, $|c_{60}| = 1/\pi = 0.318$, $c_{120} = 0$ (all even multiples vanish for 50% duty), $|c_{180}| = 1/(3\pi) = 0.106$. At 3000 rpm order 60 is $60 \cdot 50 \text{ Hz} = 3 \text{ kHz}$.

b) Each tooth contributes $\frac{1}{360} \int_{6m}^{6m+3} e^{-j2\pi k\theta/360} d\theta = \frac{3}{360} \text{sinc} \frac{k}{120} e^{-j\pi k/120} e^{-j2\pi km/60}$. The sum over $m = 0, \dots, 57$ is geometric: $|\sum_m e^{-j2\pi km/60}| = |\sin(58\pi k/60)/\sin(\pi k/60)|$ (and 58 for $k = 60m$), which gives the formula. Values: $c_0 = 58 \cdot 3/360 = 0.4833$; $|c_1| = \frac{3}{360} \cdot 0.99989 \cdot \frac{\sin(\pi/30)}{\sin(\pi/60)} = 0.008333 \cdot 0.99989 \cdot 1.9973 = 0.01664$; $|c_{59}| = 0.01077$; $|c_{60}| = \frac{58}{120} \cdot \frac{2}{\pi} = 0.3077$; $|c_{61}| = 0.01042$. Interpretation: $x_{60-2} = x_{60} - x_{\text{gap}}$, where x_{gap} are the two missing pulses (period 360°). x_{gap} has lines at *all* orders, $|c_k| \approx 2 \cdot 3/360 = 1/60$ at low orders. The gap modulates the tooth signal once per revolution: the tooth line at order 60 loses $2/60$ of its amplitude ($0.3183 \rightarrow 0.3077$) and is surrounded by sidebands at $60 \pm 1, 60 \pm 2, \dots$, like an amplitude-modulated carrier. The low orders ($k = 1, 2, \dots$) carry the reference-mark information.

c) $x(t) = \frac{1}{2j} (e^{(-1/\tau + j\omega_0)t} - e^{(-1/\tau - j\omega_0)t})$, $t \geq 0$:

$$X(f) = \frac{1}{2j} \left[\frac{1}{1/\tau + j2\pi(f - f_0)} - \frac{1}{1/\tau + j2\pi(f + f_0)} \right].$$

For $f \approx f_0 \gg 1/\tau$ the second term is negligible: $|X(f)| \approx \frac{\tau/2}{\sqrt{1 + (2\pi\tau(f - f_0))^2}}$ – a Lorentzian with peak $\tau/2$. Half power at $2\pi\tau|f - f_0| = 1$, so $\Delta f = 1/(\pi\tau)$ and $Q = \pi f_0\tau$. Knock: $\Delta f = 398 \text{ Hz}$, $Q = 16.3$; valve: $\Delta f = 796 \text{ Hz}$, $Q = 10.1$. A knock band-pass much narrower than 400 Hz would not capture more signal energy but would ring longer than the burst itself (its own impulse response lasts $\approx 1/(\pi B)$); a bandwidth of 0.5 kHz to 1 kHz matches the physics. The faster decay of the valve burst makes its peak twice as wide.

d) $f = k n/60$:

	1500 rpm	3000 rpm	6000 rpm
order 0.5 (cam)	12.5 Hz	25 Hz	50 Hz
order 2 (combustion)	50 Hz	100 Hz	200 Hz
order 60 (teeth)	1.5 kHz	3 kHz	6 kHz
6.5 kHz resonance	order 260	order 130	order 65

Crank-synchronous components have fixed orders and frequencies proportional to speed; structural resonances have a fixed frequency and a speed-dependent order. At 6000 rpm the tooth frequency even reaches the knock band.

Exercise 6.2 Numerical Fourier series and DTFT**JSLINUX**

- a) The program `code/jslinux/sheet06/fourier_series.c` samples one revolution of the crank signal `es_crank()` on a grid of $M = 36000$ points (0.01°). Implement the direct summation

$$c_k \approx \frac{1}{M} \sum_{n=0}^{M-1} x[n] e^{-j2\pi kn/M}, \quad k = 0, \dots, 1200$$

(use a rotating phasor instead of calling `cos` and `sin` for every term) and compare with your results of 6.1 b).

- b) Implement the partial sum $S_K(\theta) = c_0 + \sum_{k=1}^K 2 \text{Re}\{c_k e^{j2\pi k\theta/360}\}$ and evaluate it around the gap for $K = 60, 180, 600, 1200$. Report maximum, overshoot and rms error. Explain the Gibbs phenomenon: what converges and what does not?
- c) `code/jslinux/sheet06/dtft.c`: implement the DTFT by direct summation. For a damped sinusoid ($f_0 = 6.5 \text{ kHz}$, $f_s = 50 \text{ kHz}$, $N = 500$) and $\tau = 0.4/0.8/1.6 \text{ ms}$ the program determines

peak frequency, peak value and -3dB bandwidth. Compare with 6.1 c) (note: $X_{\text{DTFT}} \approx f_s X(f)$), and compare the finite sum with the closed-form DTFT of $a^n \sin(\Omega_0 n) u[n]$.

d) Check numerically: $|X|$ at f , $f + f_s$, $-f$ and $f_s - f$. Explain.

e) Apply the DTFT to a real knock burst of the virtual engine (both modes, noise, combustion vibration). Identify all features of the spectrum.

Solution

a)

```

1  for (k = 0; k <= KMAX; k++) {
2      /* rotating phasor instead of cos/sin per term (saves time) */
3      double cw = cos(2.0 * M_PI * k / M), sw = sin(2.0 * M_PI * k / M);
4      double pr = 1.0, pi = 0.0, re = 0.0, im = 0.0, t;
5      for (n = 0; n < M; n++) {
6          re += x[n] * pr;          /* x * exp(-j w n) */
7          im -= x[n] * pi;
8          t = pr * cw - pi * sw; pi = pr * sw + pi * cw; pr = t;
9      }
10     c[k].re = re / M; c[k].im = im / M;

```

k	$ c_k $ numeric	$ c_k $ analytic	ratio
0	0.483333	0.483333	1.0000
1	0.016642	0.016642	1.0000
2	0.016568	0.016568	1.0000
30	0.000000	0	–
31	0.000780	0.000780	1.0000
59	0.010772	0.010772	1.0000
60	0.307701	0.307700	1.0000
61	0.010419	0.010419	1.0000
120	0.000000	0	–
180	0.102571	0.102567	1.0000

The direct sum is a rectangle-rule integration; with 0.01° steps and tooth edges on the grid it is exact to 4–6 digits. The small deviation at $k = 180$ is the aliasing of the sampled pulse train (orders 180 ± 36000). Order 30 vanishes because the 58 tooth phasors $(-1)^m$ cancel; order 120 because of the 50% duty cycle.

b)

```

1  static double partial_sum(double theta, int K)
2  {
3      double s = c[0].re, w = 2.0 * M_PI * theta / 360.0;
4      int k;
5      for (k = 1; k <= K; k++)
6          s += 2.0 * (c[k].re * cos(k * w) - c[k].im * sin(k * w));
7      return s;
8  }

```

K	$\max S_K$	overshoot	$\min S_K$	rms error
60	1.1675	16.75 %	−0.3257	0.2346
180	1.1356	13.56 %	−0.1705	0.1413
600	1.0962	9.62 %	−0.0905	0.0833
1200	1.0923	9.23 %	−0.0893	0.0599

With $K = 60$ only the fundamental of the tooth signal and the gap-sidebands are present: the teeth look like a sine. Increasing K makes the ringing narrower (period $\approx 360^\circ/K$) and the

rms error falls ($\propto 1/\sqrt{K}$), but the overshoot next to every edge does not disappear; it tends to 0.0895 of the jump (Gibbs constant; 9.2% at $K = 1200$, the neighbouring edges only 3° away still contribute). The Fourier series converges in the mean-square sense, not uniformly, at a discontinuity. Consequence: band-limiting a rectangular signal (e.g. an input filter) always produces overshoot – relevant for threshold-based edge detection.

c)

```

1 static cpx dtft(const double *s, int n, double f)
2 {
3     cpx X = {0.0, 0.0};
4     double w = 2.0 * M_PI * f / FS;
5     int k;
6     for (k = 0; k < n; k++) {
7         X.re += s[k] * cos(w * k);
8         X.im -= s[k] * sin(w * k);
9     }
10    return X;
11 }

```

τ [ms]	peak [Hz]	$ X $ peak	$f_s\tau/2$	Δf_{-3dB} [Hz]	$1/(\pi\tau)$ [Hz]	Q
0.4	6490	10.00	10	798	796	8.1
0.8	6500	20.00	20	398	398	16.3
1.6	6500	39.92	40	199	199	32.7

Bandwidth and peak agree with the Lorentzian of 6.1 c) (peak of $X(f)$ is $\tau/2$, times f_s for the DTFT). Doubling τ halves the bandwidth and doubles Q (time–bandwidth product $\tau\Delta f = 1/\pi$). Small effects: the peak at 6490 Hz for $\tau = 0.4$ ms (the negative-frequency term pulls it slightly), and $39.92 < 40$ for 1.6 ms because the record of 10 ms cuts the burst after 6.25τ . Direct sum vs. closed form $X = a \sin \Omega_0 e^{-j\Omega} / (1 - 2a \cos \Omega_0 e^{-j\Omega} + a^2 e^{-2j\Omega})$, $a = e^{-1/(f_s\tau)}$: maximum difference $7.5 \cdot 10^{-5}$, consistent with the truncated tail $a^N = 3.7 \cdot 10^{-6}$.

d) Output: $|X| = 7.6224$ at 6000 Hz, 56 000 Hz, -6000 Hz and 44 000 Hz; $X(6000) = 7.1174 - 2.7283j$, $X(-6000) = 7.1174 + 2.7283j$. Periodicity: $e^{-j2\pi(f+f_s)n/f_s} = e^{-j2\pi fn/f_s}$ for integer n – this is aliasing seen from the frequency domain. Conjugate symmetry $X(-f) = X^*(f)$ holds for real $x[n]$; together: $|X(f_s - f)| = |X(f)|$, so $[0, f_s/2]$ contains all information.

e) Knock burst of amplitude 1.50 V: mode 1 peak at 6470 Hz, $|X| = 28.5$ (expected $1.5 \cdot 20 = 30$), bandwidth 410 Hz; mode 2 at 10 410 Hz, $|X| = 13.9$ (expected $0.75 \cdot 20 = 15$), bandwidth 380 Hz. In addition: a broad low-frequency hump (0 Hz to 500 Hz) from the combustion vibration and the noise floor of the white sensor noise. The deviations from the ideal values (peak positions ± 90 Hz, heights -5%) are caused by the noise and the overlap of the two modes and the low-frequency part in a single, random burst.

Exercise 6.3 Engine orders

JSLINUX

code/jslinux/sheet06/orders.c records the knock-sensor signal of the virtual engine (knock probability 0.2) at constant 1500, 3000 and 6000 rpm, each over exactly 8 engine cycles (16 revolutions, $f_s = 50$ kHz), removes the mean and evaluates the amplitude spectrum $A(f) = 2|X(f)|/N$ by direct summation on 0 Hz to 1000 Hz (1 Hz steps) and 4 kHz to 12 kHz (10 Hz steps).

- Implement `amp_spectrum()`. Why is the record chosen as an integer number of revolutions?
- The combustion vibration of the model is a half-sine $0.2 \text{ V} \cdot \sin(\pi\varphi/60^\circ)$ for $0 \leq \varphi < 60^\circ$ after every TDC (period 180°). Compute its Fourier series and the amplitudes of orders 2, 4, 6, 8; enter your formula in `halfsine_order_amp()`.
- Run the program. Compare the low-frequency lines (frequency and amplitude) at the three speeds with b). For the high band, the program smooths $|X|^2$ over ± 150 Hz and reports the

peak in three bands – tabulate frequency and order of the peaks.

- d) Explain the fine structure of the high-frequency spectrum: why does the valve resonance appear as a comb of lines with a spacing of order 2 under a Lorentzian envelope, while knock gives a smooth hump?
- e) An ECU wants to monitor a crank-synchronous vibration and the knock resonance with fixed band-pass filters. Which of the two filters must be adapted to the engine speed?

Solution

a)

```

1 static void amp_spectrum(const double *s, int n, const double *f, double *A, int nf)
2 {
3     int i, k;
4     for (i = 0; i < nf; i++) {
5         double w = 2.0 * M_PI * f[i] / FS, cw = cos(w), sw = sin(w);
6         double pr = 1.0, pi = 0.0, re = 0.0, im = 0.0, t;
7         for (k = 0; k < n; k++) {
8             re += s[k] * pr;
9             im -= s[k] * pi;
10            t = pr * cw - pi * sw; pi = pr * sw + pi * cw; pr = t;
11        }
12        A[i] = 2.0 * sqrt(re * re + im * im) / n;
13    }
14 }
```

With 16 revolutions every order k has $16k$ complete periods in the record: the components are “bin-centred” and $2|X|/N$ is their exact amplitude without leakage into neighbouring frequencies (Sheet 8). The mean is removed because the half-sines have a DC value of 42 mV whose spectral main lobe would otherwise dominate the lowest frequencies.

b) With $w = 60^\circ$, $P = 180^\circ$, $A = 0.2$ V and $\beta = 2\pi m/P$: $\int_0^w \sin(\pi\varphi/w)e^{-j\beta\varphi} d\varphi = \frac{\pi/w(1+e^{-j\beta w})}{(\pi/w)^2 - \beta^2}$, hence with $r = 2mw/P = 2m/3$

$$|c_m| = \frac{Aw}{P} \frac{2}{\pi} \frac{|\cos(\pi r/2)|}{|1 - r^2|}, \quad A_{2m} = 2|c_m|.$$

Harmonic m of the 180° period is engine order $2m$: $A_2 = 76.4$ mV, $A_4 = 54.6$ mV, $A_6 = 28.3$ mV, $A_8 = 6.9$ mV ($c_0 = Aw/P \cdot 2/\pi = 42.4$ mV).

c) Low-frequency lines (amplitudes in mV):

	order 2	order 4	order 6	order 8
theory	76.4	54.6	28.3	6.9
1500 rpm (50/100/150/200 Hz)	76.6	54.4	28.5	6.6
3000 rpm (100/200/300/400 Hz)	76.3	53.7	28.4	6.3
6000 rpm (200/400/600/800 Hz)	76.0	53.0	29.4	5.6

The strongest line is always order 2 (50, 100, 200 Hz); the amplitudes are speed-independent and agree with b) within a few percent (deviations: 1 % speed ripple, random knock bursts and noise). High band (peak of the smoothed $|X|^2$):

band	1500 rpm	3000 rpm	6000 rpm
5.5 kHz to 7.5 kHz (knock 1)	6450 Hz / order 258	6460 Hz / 129	6440 Hz / 64
7.5 kHz to 8.5 kHz (valve)	8050 Hz / 322	7950 Hz / 159	7860 Hz / 79
9.5 kHz to 11.5 kHz (knock 2)	10 520 Hz / 421	10 480 Hz / 210	10 580 Hz / 106

The resonances stay at their frequencies (within the ± 150 Hz smoothing and the random variation of the few knock events) while their order halves with every doubling of the speed; the low-frequency lines keep their order and double their frequency.

- d) The valve impact excites the same damped 8 kHz oscillation with the same phase every 180° : a periodic repetition of one burst. Periodic repetition with period T_{180} samples the burst spectrum (Lorentzian, 796 Hz wide) at multiples of $1/T_{180} = \text{order } 2$ (50, 100 or 200 Hz spacing) – a line spectrum under a fixed envelope. The knock bursts occur randomly (probability 0.2) with random amplitude and phase: no periodicity, hence no lines but a continuous Lorentzian hump at 6.5 kHz/10.5 kHz. (At 6000 rpm the valve envelope is sampled by only a few lines 200 Hz apart, which together with the knock skirt shifts the smoothed maximum to 7.86 kHz.)
- e) The filter for the crank-synchronous vibration must follow the speed (order tracking, or processing in the angle domain); the knock band-pass is fixed in frequency (but its measuring window is defined in crank angle).

Exercise 6.4 Order tracking during a run-up

CHALLENGE

code/jslinux/sheet06/order_track.c: the engine accelerates from 1500 rpm to 6000 rpm within 1 s; the knock-sensor signal (no knock) is sampled at 50 kHz, crank edges are time-stamped as in Exercise 4.4.

- The program computes the amplitude spectrum of the time signal on 0 Hz to 500 Hz. Describe it. Why can no order amplitude be read from it?
- Resample the signal onto a 0.5° grid (code of 4.4) and compute the “order spectrum” $A(O) = 2|\sum_m x_a[m]e^{-j2\pi Om\Delta\theta/360^\circ}|/N_a$ for $O = 0 \dots 10$ over an integer number of revolutions. Compare the order amplitudes with Exercise 6.3.
- The angle grid has a “sampling rate” of $360^\circ/\Delta\theta = 720$ samples per revolution. Which orders can be represented, and which signal components alias in the angle domain at 1500 rpm? Why does this not disturb the low orders here?

Solution

- a) The 2nd order sweeps from 50 to 200 Hz, the 4th from 100 to 400 Hz, ...: the spectrum is a broad smeared plateau with a maximum of only 11.9 mV (instead of 76 mV); overlapping sweeps add up between 100 and 200 Hz. The energy of each order is distributed over the whole frequency range it passes; the DTFT assumes stationarity.

b)

```

1  for (k = 2; k < ne; k++) {
2      double d1 = te[k] - te[k - 1], d0 = te[k - 1] - te[k - 2];
3      if (kref < 0) { if (d1 > 2.0 * d0) { kref = k; the[k] = 0.0; } }
4      else the[k] = the[k - 1] + (d1 > 2.0 * d0 ? 18.0 : 6.0);
5  }
6  {
7      int ke = kref, ks = 0;
8      while (na < MAXA) {
9          double th = na * DTH, tq;
10         if (th > the[ne - 1]) break;
11         while (ke < ne - 2 && the[ke + 1] < th) ke++;
12         tq = te[ke] + (te[ke + 1] - te[ke]) * (th - the[ke]) / (the[ke + 1] - the[ke]);
13         if (tq >= ts[NS - 1]) break;
14         while (ks < NS - 2 && ts[ks + 1] < tq) ks++;
15         xa[na++] = xs[ks] + (xs[ks + 1] - xs[ks]) * (tq - ts[ks]) / (ts[ks + 1] - ts[ks]);
16     }
17 }
18 nrev = (int)(na * DTH / 360.0);          /* whole revolutions only */
19 na = (int)(nrev * 360.0 / DTH);
20 remove_mean(xa, na);
21 for (i = 0; i < NORD; i++) ord[i] = 0.01 * i;
22 /* order O = cycles per revolution; angle step DTH -> w = 2 pi O DTH/360 */
23 amp_spec(xa, na, ord, DTH / 360.0, ao, NORD);

```

Result: 61 revolutions, 43920 angle samples; order amplitudes $A_2 = 75.5 \text{ mV}$, $A_4 = 53.7 \text{ mV}$, $A_6 = 29.6 \text{ mV}$, $A_8 = 5.6 \text{ mV}$ – sharp lines with the same amplitudes as at constant speed (6.3 c), although the speed quadrupled. Angle-domain resampling turns the non-stationary time signal into a stationary angle signal (*computed order tracking*).

c) Orders up to $O < 360$ are represented. The time signal contains components up to $f_s/2 = 25 \text{ kHz}$, i.e. up to order $25000/25 = 1000$ at 1500 rpm: the valve resonance lies at order 320 (still below 360), but a knock mode at 10.5 kHz would appear at order 420 and alias to $720 - 420 = 300$, and the broadband sensor noise above order 360 folds back into $[0, 360]$. Aliasing in the angle domain only matters if strong content lies near $720k \pm O$ for the orders O of interest; for $O = 2 \dots 8$ this is only white noise, which is spread thinly over all orders (the linear interpolation additionally acts as a weak low-pass). For a clean order analysis the time signal must be low-pass filtered (or the grid refined) before resampling – the anti-aliasing rule of Sheet 5 in the angle domain.

Deliverables

6.1: derivations, coefficient values, Lorentzian bandwidths, order table. 6.2: completed programs, coefficient table, Gibbs table with explanation, DTFT table and the checks d)/e). 6.3: completed program, order amplitudes (theory vs. measurement) and resonance table with interpretation. 6.4 (optional): order spectrum vs. time spectrum of the run-up.